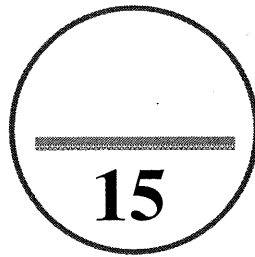


ANSWER SHEET

Initials and surname	Memo.							
Student nr.								
Degree	B.Eng ()							

Final Mark:



Write final answers in the table below.

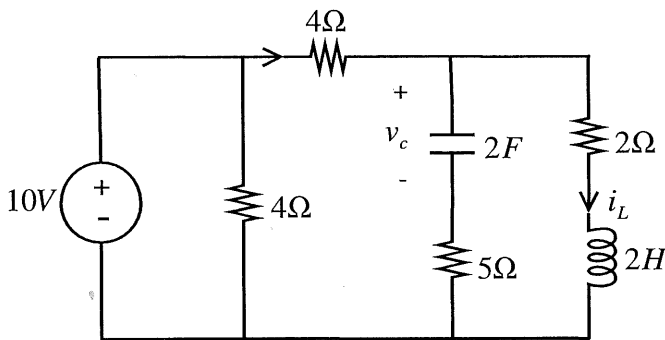
Question	Answer
1	$v(t) = \cancel{20} \sqrt{-2 \sin 20t} \quad \checkmark \quad 2 \cos(20t + 90^\circ)$
2	$W_{\text{ind}} = 2.777 \text{ J} \quad \checkmark$ $W_{\text{cap}} = 11.111 \text{ J} \quad \checkmark$ (-1) for incorrect units.
3	a) $5 \cos(\omega t - 90^\circ) \quad \checkmark$ b) $10 \cos(\omega t - 130^\circ) \quad \checkmark$ - $5 \cos(\omega t + 90^\circ)$ No $\frac{1}{2}$ marks
4	$\underline{V_1}$ leads $\underline{V_2}$ with $\underline{30^\circ}$ degrees
5	$v(t) = 29.595 \cos(\omega t - 133.36^\circ) \quad \checkmark$ No marks for polar
6	$0.82 + 5.03j \quad \checkmark$ No marks for polar
7	$3.23 \angle -72.33^\circ \quad \checkmark$ No marks for rectangular.

EBN 111 Class test#3: 21 May 2012 (15 marks, 30 minutes)

Question 1 (2): A current source $i(t) = 10\cos(20t)$ A is connected across a 10 mH inductor. Calculate the voltage $v(t)$ across the inductor.

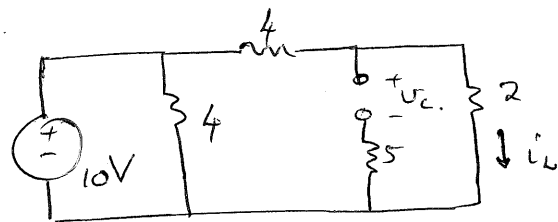
$$\begin{aligned}
 v &= L \frac{di}{dt} \\
 &= 10 \times 10^{-3} \frac{d}{dt} 10 \cos(20t) \\
 &= 100 \times 10^{-3} (-20 \sin(20t)) \\
 &= -2 \sin 20t \text{ V.}
 \end{aligned}$$

Question 2 (2): Calculate the stored energy in the inductor and capacitor in the steady state.



$$i_L = \frac{10}{4+2} = 1.667 \text{ A.}$$

$$v_c = 2 \times i_L = 3.333 \text{ V}$$



$$w_L = \frac{1}{2} L i^2 = 2.777 \text{ J.}$$

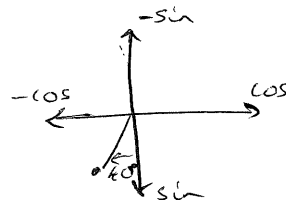
$$w_C = \frac{1}{2} C v^2 = 11.111 \text{ J}$$

Question 3 (2): Write the following functions in cosine-form [$A\cos(\omega t + \phi)$, $A > 0$, $|\phi| \leq 180^\circ$]:

- a) $-5\sin(\omega t + 180^\circ)$
- b) $10\sin(\omega t - 40^\circ)$

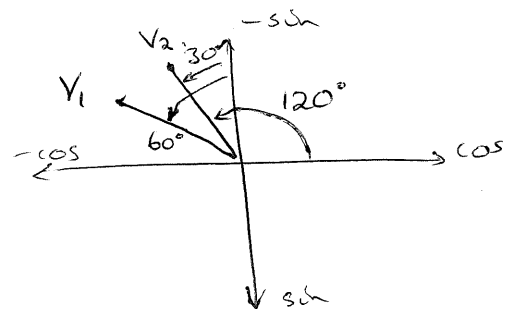
$$\begin{aligned}
 \text{a) } &-5\sin(\omega t + 180^\circ) \\
 &= 5\sin(\omega t) \\
 &= 5\cos(\omega t - 90^\circ).
 \end{aligned}$$

$$\begin{aligned}
 \text{b) } &10\sin(\omega t - 40^\circ) \\
 &= 10\cos(\omega t - 130^\circ).
 \end{aligned}$$



Question 4 (2): Which of the following sinusoids is leading and with how many degrees?

$$v_1(t) = -20\sin(\omega t + 60^\circ); \quad v_2(t) = 10\cos(\omega t + 120^\circ)$$



V_1 leads by 30°

Question 5 (2): Use phasors to write the following as a single cosine expression [$A \cos(\omega t + \phi)$, $A > 0$, $|\phi| \leq 180^\circ$]: $v(t) = 20 \sin(\omega t - 50^\circ) - 10 \cos(\omega t + 60^\circ)$

$$\cancel{V} \quad v(t) = 20 \cos(\omega t + \cancel{40^\circ} - 140^\circ) - 10 \cos(\omega t + 60^\circ)$$

$$\begin{aligned} \bar{V} &= 20 \angle -140^\circ - 10 \angle 60^\circ \\ &= 29.595 \angle -133.36^\circ \end{aligned}$$

$$v(t) = 29.595 \cos(\omega t - 133.36^\circ)$$

Question 6 (2): Evaluate the following and write in rectangular form:

$$\begin{aligned} &\frac{10 + j20}{(3 - j4)} + 3 \angle 20^\circ \\ &= \frac{22.36 \angle 63.43^\circ}{5 \angle -53.13^\circ} \\ &\quad + 2.82 + 1.03j \\ &= -2 + 4j \\ &\quad + 2.82 + 1.03j \\ &= 0.82 + 5.03j \end{aligned}$$

Question 7 (3): Evaluate the following and write in polar form:

$$\begin{aligned} &\frac{(14 + j9) + 10e^{j30^\circ}}{(-2 + j8)} \\ &= \frac{14 + j9 + 10 \angle 30^\circ}{8.25 \angle 104.04^\circ} \\ &= \frac{22.66 + 14j}{8.25 \angle 104.04^\circ} \\ &= \cancel{0.98} / 3.23 \angle -72.33^\circ \end{aligned}$$